An open book by TopLocalAI

Use AI to Make Better Scientific Figures

From Research Ideas to Paper Figures

Practical workflows with SciDraw AI

English edition · Working draft v0.1.0

2026-10-29

Preface and Chapters 1–2; later chapters remain on the writing roadmap.

Version and use

This open book is maintained by TopLocalAI in LaTeX, with Chinese and English editions. The current draft contains the preface, Chapters 1 and 2, a complete chapter plan, exercises, and asset provenance.

SciDraw AI provides the practical workflow used throughout the book. In this edition, Codex assists with the example figures: the data plot is rendered from a published simulated CSV by a reproducible script, and the mechanism illustration is an actual output of Codex's built-in image generation. Neither is presented as a SciDraw AI generation result. Interface screenshots show a local development version; public availability of the guide must be checked separately.

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`https://github.com/TopLocalAI/ai-scientific-figures-book`

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Preface: Bring attention back to the research

We often leave figures until the final stages of writing a paper. The data have been analyzed, the methods are documented, and the conclusions seem clear. All that remains is to make the figures look good. Once we begin, problems appear: one image cannot hold everything, two arrows invite different interpretations, a summary hides individual variation, and text that looked clear on a monitor becomes unreadable at its final size.

These difficulties are closely connected to the research. Making a figure requires us to decide again what matters most, which elements are evidence, which are interpretation, and in what order the reader should encounter them. Sometimes a figure is unclear because its layout has not been resolved. Sometimes the question it should answer has not yet been stated precisely.

AI lowers the effort needed to begin. We can describe a scene in words, suggest a composition with a sketch, explain a revision using an existing image, or give data to a visualization tool and explore ways to present them. Tasks that once required many separate operations can now start with a short instruction. That creates more room to compare alternatives and refine the explanation.

A polished image can also conceal mistakes. An incorrect relationship can look convincing. A tidy layout can distract from a missing experimental condition. An unsupported conclusion can acquire a professional-looking visual form. We therefore need to move from being able to generate a figure to being able to explain, inspect, and deliver it.

This book follows that process. You will begin with a drawing brief that states the purpose of the figure and the question it should answer. You will then choose a workflow for a data plot, a methods diagram, a mechanism illustration, an experimental image, or a graphical abstract. We will discuss composition, color, labels, and legends, as well as revision, export, and preservation of source materials. Each part explains what AI can assist with and what must be decided from the research record.

SciDraw AI is the practical platform used throughout the book. A common platform lets readers continue working on the same example without repeatedly learning a new interface. Instructions are based on actual interfaces and available operations. Where a workflow requires a download, an upload, or a manual addition, that step is stated directly. Features enter a tutorial after they have been implemented and checked.

You do not need to master illustration before starting. More useful preparation is a study you understand, a complete set of source materials, and a willingness to inspect each important element. Experienced readers can enter at a specific problem: retaining the study design in a data plot, reducing ambiguity in a mechanism diagram, or arranging several panels into a readable argument.

Exercises distinguish simulated materials from real research. Simulated data allow us to share a reproducible task and discuss mistakes without presenting a teaching result as a scientific discovery. Real cases require reliable sources and any necessary permission. The aim is a method that transfers to your own work: decide what to communicate, use the tools, and preserve important decisions together with their basis.

In the next chapter, focus on one concrete question: what should this figure help the reader understand?

Chapter 1

Build your AI scientific-figure workflow

Suppose you have a table containing two measurements for each of eight objects. You want to turn it into a figure for a paper. If you enter "Make a professional data plot," the tool may return an apparently complete result. Yet that instruction does not tell it whether the columns belong to the same objects, or whether your question concerns average differences, individual changes, or agreement between measurements.

A small omission can change the study design communicated by the figure. This chapter addresses that problem by organizing the work before operating the tool.

1.1 Give the figure a testable task

A scientific figure should answer a principal question. It might explain the order of a process, relationships among objects, or changes between two measurements of the same object. The question should be specific enough to influence the chart type, layout, or labels.

"Show the experimental results" is too broad. "Show each object's change and make it possible to see whether the changes are consistent" is more useful. It requires individual correspondence, direct comparison between measurements, and space for an object that moves against the majority.

Identifying the audience and the message before designing a figure is a basic practice in scientific visualization. Rougier and colleagues present these as separate rules; this book places them at the start of every brief.[1]

Write your first sentence:

> After viewing this figure, the reader should understand _____
> and be able to locate _____ in the figure.

The first blank states the purpose; the second identifies something the reader can check. For example: "understand the changes in eight objects and locate the two observations for each object." This statement does not require the result to match an expectation. It requires the figure to accommodate different outcomes.

1.2 Identify the material you are handling

We use three operational categories. A paper, or even a single multi-panel figure, can contain all three.

Material	Main content	AI assistance	Author review
Numbers and tables	Quantities, distributions, changes	Plot proposals, chart creation, styling	Values, units, groups, pairing, statistics
Text, sketches, confirmed relationships	Structure, methods, mechanisms	Layout candidates, illustrations, revisions	Objects, counts, direction, conditions, evidence
Acquired experimental images	Observed form, signal, spatial information	Presentation planning, descriptions, checks	Originals, calibration, processing, comparability

The category affects the tool you choose. Use the scientific-visualization data workflow when values and their correspondence must be preserved. Use the drawing workflow to explain a confirmed process. For experimental images, keep the original files and then work on presentation and panel assembly.

A graphical abstract often combines categories. Icons may explain a method while a small data plot summarizes a result. Each part still needs the treatment appropriate to its source. A schematic that explains an apparatus and an image actually acquired by that apparatus play different evidential roles.

1.3 Organize the work into six stages

You do not need a complex project system. Begin with six stages: define the task, prepare materials, create a draft, review and revise, organize the explanation, and check and deliver.

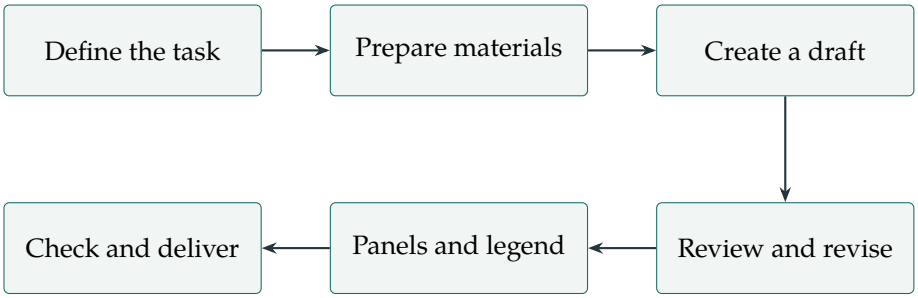


Figure 1.1. The book's workflow. A task may omit stages such as panel assembly. When a check identifies a problem, return to the stage that can resolve it.

Define the task. State the use, audience, and main question. The useful output at this stage is an editable drawing brief.

Prepare materials. Locate the data, original images, supporting relationships, and necessary methods. Confirm that they belong to compatible versions and list missing information. Similar filenames do not make two files interchangeable.

Create a draft. Provide the brief and materials to the appropriate tool. Save the instruction actually submitted, so that later you can distinguish an unclear request from an incorrectly executed one.

Review and revise. Check scientific content before presentation. Ask for a specific change, such as "increase the axis-label size while retaining every point and pairing." After each revision, inspect the result again, including elements that were not meant to change.

Organize the explanation. For several images, decide panel order and numbering. When writing a legend, add information that cannot be inferred from the image alone. Compare an AI-generated draft with the experimental record.

Check and deliver. Inspect the figure at its final display size, verify the technical properties, and retain the legend and source materials. A colleague receiving the file should be able to identify the data or original-image version behind it.

AI has a different role in each stage. A first draft makes a possible explanation visible. A revision carries out a defined change. A legend draft supplies

text to inspect. Adoption depends on agreement with the brief and source materials.

1.4 Choose a route in SciDraw AI

Open the scientific-figure workflow guide and select the current task. Its four routes cover mechanisms and workflows, data visualization, assembly of existing images, and graphical abstracts. Each route provides steps, materials to bring, outputs to save, and checks before continuing.

Start at "Prepare your brief" when beginning a figure. If you already have an image and need a legend, enter at that stage. The guide supplies a route; the work itself takes place in the existing drawing, visualization, panel, legend, and checking tools.

This version stores the brief and progress in the current browser. Copy the brief or download it as text. Browser storage does not replace a source-material folder: changing browsers, clearing site data, or using an environment that blocks storage may remove access to the record. Download anything you need to retain.

"Next step" only changes the displayed step. Mark a step complete after checking the work. You can also undo that status. Progress should represent a decision you made, rather than a count of pages visited.

When a tool opens from the guide, the task and step accompany the link. Relevant pages provide a return control, and a drawing conversation can be reopened through its reference. Images currently move between tools through downloading and uploading. Save the current figure, then select that file in the panel, legend, or checking tool. A version in the filename reduces the chance of uploading the wrong image.

1.5 Exercise: plan a plot for paired data

Download `paired-example.csv` and read its data dictionary. It contains eight fictional objects with measurements named `before` and `after`. Values use arbitrary units, abbreviated a.u. They were constructed for teaching and do not come from an experiment.

Consider three rows:

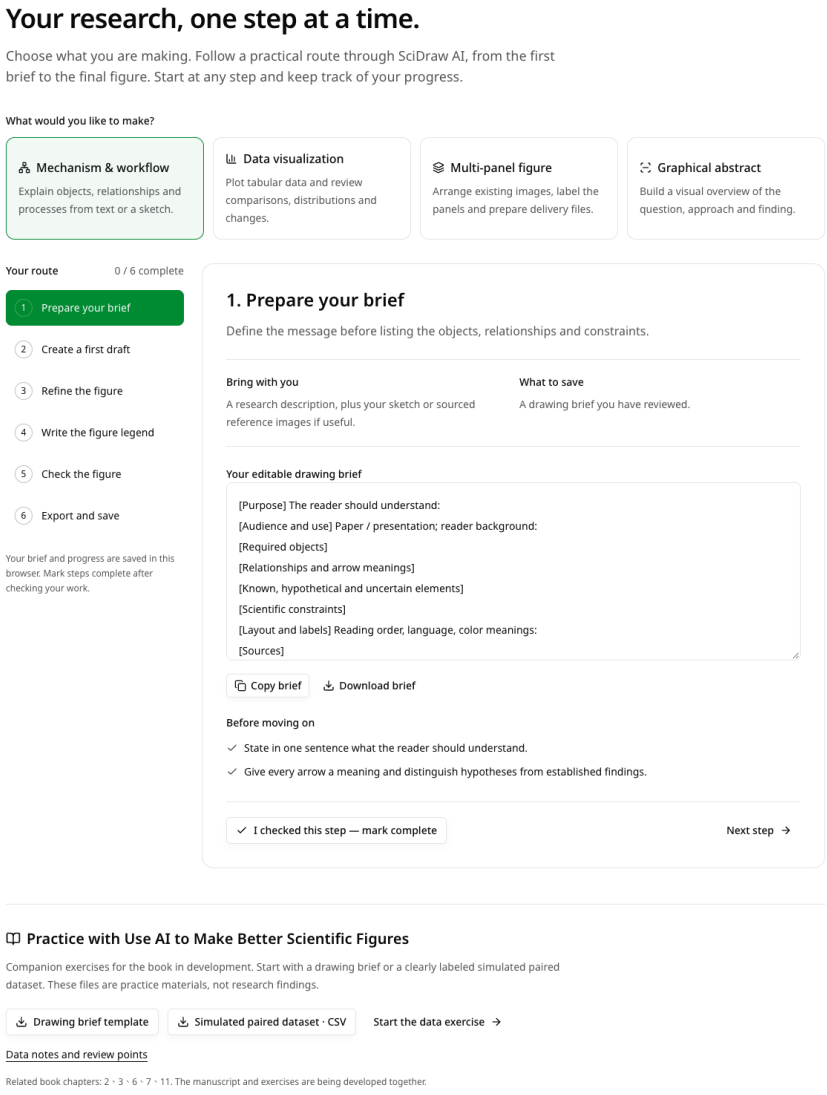


Figure 1.2. The English development interface of the SciDraw AI figure-workflow guide, captured on October 29, 2026. This local preview illustrates task navigation; it does not establish that the guide has been deployed publicly.

sample_id	before	after
SIM01	12	16
SIM02	15	18
SIM04	18	17

SIM01 and SIM02 increase; SIM04 decreases. If the task is to show individual changes, that third row must remain visible. Summaries of the two measurements alone do not retain individual correspondence. Weissgerber and colleagues discuss information that can be obscured when paired data are reduced to summary plots, and emphasize communicating study design.[2]

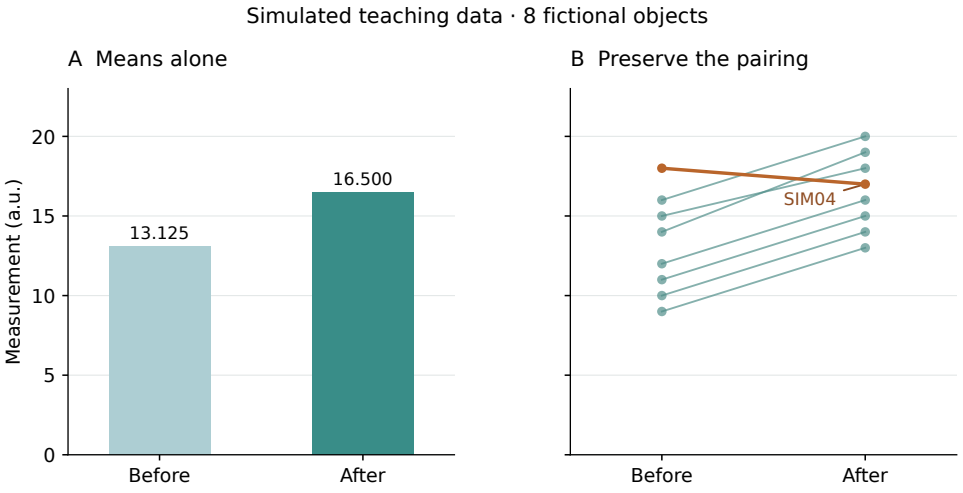


Figure 1.3. Two views of the same simulated dataset. The left panel retains only the two means. The right panel preserves all eight pairs, including SIM04, which decreases from 18 to 17. Codex wrote the plotting script; the script reads the CSV without generating or altering observations. No significance test is performed.

Select the data route in the guide and replace the default brief with the following:

```
Use: practice data presentation; report no research finding.
Purpose: show the changes in eight fictional objects.
Source: paired-example.csv; sample_id identifies each pair.
Display: paired points and lines, before on the left, after on
↪  the right.
```

```
Constraints: retain all eight objects, all sixteen values, and
↪  the original pairing.
Y-axis: measurement in a.u.; do not change any value.
Label: state "Simulated teaching data" in the title or legend.
Analysis: no significance test, causal claim, or
↪  treatment-effect claim.
Review: compare every endpoint with its row; SIM04 must
↪  decrease from 18 to 17.
```

This is a workable starting point. A paired point-and-line plot is a candidate for this example, not a universal rule for paired data. Many objects, serious overlap, or a different question can justify another presentation.

Complete three tasks now: save the brief, confirm the data dictionary, and list acceptance criteria. This edition also provides the deterministic Codex-authored plotting script and its output as a reference. Chapter 5 will add the actual SciDraw AI visualization session. Do not mark an unperformed generation step complete just to increase the progress count.

1.6 Keep a small archive for each figure

An ordinary folder can support handover. The following is a suggested structure, not a project automatically created by the platform:

```
paired-figure/
  source/    Original CSV and data dictionary
  brief/     Brief and instructions actually submitted
  drafts/    Numbered candidates and revision notes
  final/     Final figure, legend, and delivery notes
```

You do not need to record every mouse movement. Prioritize changes that affect interpretation: a transformed axis, filtered data, a different panel order, or a hypothetical relationship made definite. A colleague comparing two versions should be able to discover why they differ.

Record AI use concretely: the function used, any visible model or settings, the actual input, the selected output, and the review and revisions. Before submission, check the target journal's current requirements. A successful technical check does not establish scientific correctness or compliance with an editorial policy.

1.7 Exercise and acceptance criteria

Choose a figure you intend to make. Write its one-sentence purpose, identify the principal material category, and select a route in the guide. Complete and download the brief, then place it with the source materials.

Ask a colleague who has not watched your work to read the brief. They should be able to say what the figure must communicate, where the input is, what must remain unchanged, and what counts as completion. If their answer is only "clear, professional, and attractive," the task needs more detail.

The next chapter turns this short description into an executable and reviewable drawing brief.

Chapter 2

Turn research content into a drawing brief

> R, K, P, and I are abstract teaching symbols. They do not represent real molecules or a published mechanism. The included illustration is an actual Codex image-generation output, reviewed against these fictional requirements.

Researchers can often explain a project to a colleague in a few minutes yet struggle to tell a drawing tool what to show. A conversation allows us to supply missing assumptions and rely on shared background. A drawing instruction must make those assumptions explicit.

"Draw a pathway showing upstream activation, downstream effects, and an inhibitor" sounds like a task. But what is upstream? Which object is activated? Is the downstream effect observed or proposed? Where does the inhibitor act? Without answers, the generator must fill in structure.

A drawing brief turns the author's knowledge into instructions that can be executed and checked.

2.1 Give the brief and the prompt different jobs

The brief records what you intend to communicate and the basis for it. A prompt selects relevant parts of that brief for a particular operation. They may be nearly identical, or differ substantially across stages.

Initial generation needs objects, relationships, and layout. A color revision may need only a new color assignment and the elements that must remain unchanged. A legend requires the image and an explanation of its context. The

brief keeps these local operations aligned with the same purpose.

There is no need to force every requirement into one universal prompt. Write the brief first, then select information for the current operation. If a revision changes the scientific content, update the brief as well, so that an old specification does not continue to drive a new image.

2.2 Build the brief around seven questions

The companion template contains ten fields. These seven questions organize the thinking behind them.

Who will read it, and where? A specialist, an interdisciplinary reviewer, and a student may need different figures from the same study. Use also determines viewing size. "For colleagues familiar with the basic method, in the results section of a paper" is more informative than "journal style."

What is the main question? State the figure's central task in one sentence. If the sentence accumulates several separate aims, consider panels with distinct jobs.

Which materials support it? Identify filenames, worksheets, records, or locations in references. The aim is to support important statements, not to maximize the volume of uploaded material. Avoid mixing incompatible result versions.

Which objects and relationships are required? A list of objects is insufficient without correspondence, direction, grouping, or spatial placement. Two measurements from the same object constitute a relationship, just as an arrow does in a mechanism diagram.

What is observation, and what is interpretation? Keep observations, inferences, and hypotheses distinct. State uncertainty in the brief, express it with labels or an agreed line style, and explain the convention in the legend. A tool should not decide an unsupported relationship.

How should it be presented? Now choose chart type, reading direction, panels, color roles, and label language. Give constraints that help explain the content. Adjectives alone cannot specify a layout.

How will completion be recognized? Write criteria such as correct object counts, one-to-one correspondence with the relationship table, preservation of every value, and readable text at final size. Connect review criteria to the source materials.

Mark unresolved information as "to be confirmed." It becomes a task for the author rather than an invitation for a polished draft to settle the matter.

2.3 Replace a noun list with a relationship table

Consider a deliberately fictional model with four symbols: R, K, P, and I. For this exercise, R activates K; K has a hypothetical influence on P; and I inhibits K. These are teaching assumptions, not biological claims.

"R, K, P, I; mechanism diagram; blue-green; simple" does not determine the connections. A relationship table does.

From	To	Meaning in this model	Visual convention
R	K	Given activation	Solid arrow toward K
K	P	Hypothetical influence	Dashed arrow toward P
I	K	Given inhibition	Line ending in a T-bar at K

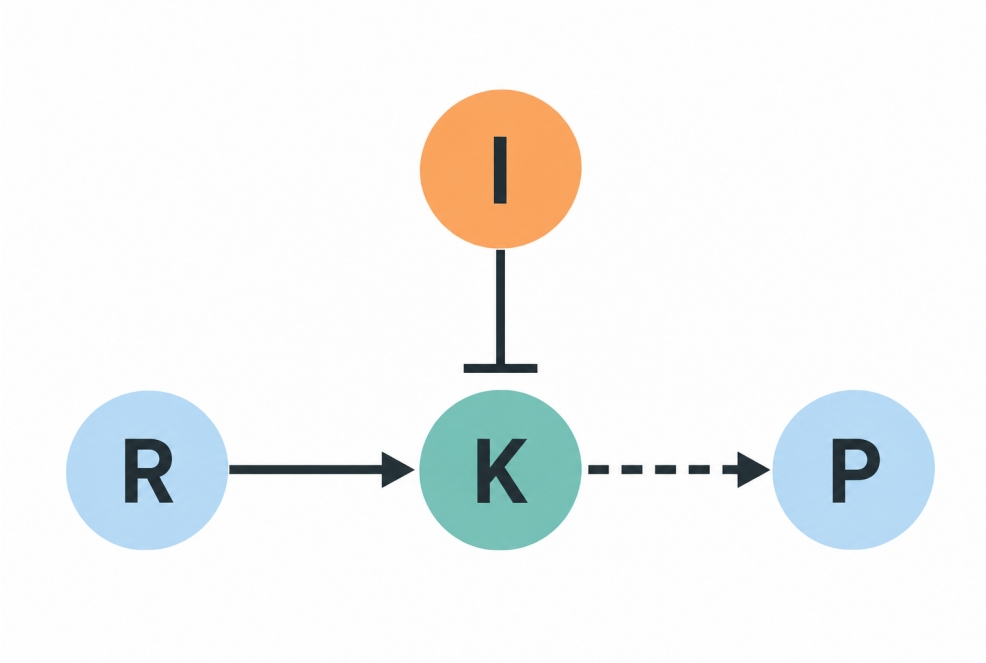


Figure 2.1. An actual candidate from Codex's built-in image generation. A solid arrow, dashed arrow, and T-bar express the three stipulated relationships. All four symbols and relationships are fictional. This output was generated in Codex, not in SciDraw AI.

Line styles and arrowheads have no universal meaning outside context. Their meaning depends on disciplinary convention, the definitions in the figure, and its legend. The conventions here distinguish three relationships. A

real project should use forms its audience understands and add explicit labels where ambiguity remains.

Review the result row by row: locate R and K and inspect the direction; check that the K-to-P connection remains dashed; then confirm that I's inhibitory endpoint is K. "Does this look right?" becomes a set of locatable questions.

2.4 Write an executable instruction for the mechanism

Select the mechanism-and-workflow route in the SciDraw AI guide, replace its default brief, and download a copy.

```
[Audience and use] Scientific-figure teaching; no biology
↪ background required.
[Purpose] Distinguish given, hypothetical, and inhibitory
↪ relationships.
[Source] Only the fictional model specified here.
[Objects] R, K, P, I; each appears once.
[Relationships]
R activates K: a solid arrow from R to K.
K may influence P: a dashed arrow from K to P.
I inhibits K: a line ending with a perpendicular T-bar next to
↪ K.
[Layout] R, K, P from left to right; I above K.
[Visual language] White background; simple two-dimensional
↪ shapes; identify every object by its letter.
[Constraints] Add no objects, relationships, molecular forms,
↪ or experimental results.
[Legend] Explain the three line conventions and identify the
↪ fictional teaching model.
[Review] Four objects and three relationships; check direction,
↪ endpoints, and labels.
```

This instruction provides concrete quality criteria: identifiable objects, accurate relationships, and a hypothesis that remains distinguishable from a given relationship. Color can be refined later; the relationships must hold first.

In the mechanism tool, place the instruction in the generation field. Review a candidate against the relationship table before requesting further changes. If an extra object appears, identify it and restate the requirement for exactly four objects, rather than asking only for a simpler design.

For this edition, Codex's built-in image generator produced Figure 2.1. The complete executed prompt and review are included in the repository. The four

labels and three relationship types match the brief. Soft tonal variation remains inside the nodes, so this is a checked teaching candidate, not proof that every style constraint was followed exactly. It is not a SciDraw AI output.

The executed generation prompt uses labels alone, with the explanatory words moved to the caption. This is an intentional layout choice recorded in the provenance. Outputs can vary between runs; an instruction does not guarantee that all requirements will be met automatically.

2.5 For a data plot, specify constraints by column

A data brief must connect study design to table structure. Three columns in the first exercise already illustrate several possible ambiguities.

`sample_id` identifies an object. `before` and `after` contain its two measurements. Eight rows represent eight fictional objects and sixteen observations. State object and measurement counts separately. In real research, also explain experimental units, repeated measurements, and technical replicates.

```
Read the uploaded paired-example.csv.
sample_id is the pairing key. before and after are two
↪  measurements in a.u.
Create a paired point-and-line plot, with before then after on
↪  the x-axis.
Connect the two observations belonging to each object.
Keep all eight objects and the sixteen original values.
Do not treat the columns as independent samples or connect
↪  different objects.
Title: "Simulated teaching data: individual changes".
Show the measurements without significance tests or error bars.
Do not remove SIM04 or change its decrease into an increase.
```

The final line is a check against the file: SIM04 really does decrease from 18 to 17 in this exercise. It does not prescribe a desired finding for real research. Define checks from your actual data, rather than forcing the figure to match an expected conclusion.

“No error bars” is also specific to this exercise. When a different plot needs them, define whether they represent standard deviation, standard error, a confidence interval, or another quantity, and identify how it is calculated. “Add standard error bars” cannot be used as a vague synonym for any uncertainty interval.

2.6 When using a PDF helper, establish what was read

A paper's information is distributed across its abstract, body, figures, methods, and supplementary materials. How text is extracted affects the content available to a visual-prompt helper.

The SciDraw AI PDF helper examined for this edition reads up to the first five pages and uses a limited extract from the abstract or beginning of the document. It can assist with a research overview, but this does not establish that the entire paper, every figure, and the supplements have been interpreted.

Before using it, prepare author-confirmed points: the research question, main method, result to show, and relevant conditions. Compare the helper's prompt with that list, restore omitted information, and only then proceed to drawing. A key result appearing later in a paper may be absent from an input limited to its opening pages.

Record the input scope in the brief. For example: "Author-prepared abstract and result notes; PDF assistance used for drafting; key statements checked against the original." This is more accurate than "automatically generated from the full paper" and makes future reconstruction easier.

2.7 Turn vague feedback into reviewable requests

Repeated generation can become a search for a more appealing image without a clear purpose. Give each operation a defined job.

Vague feedback	A more concrete request
More professional	Align R, K, and P horizontally; use one label size; retain all objects and relationships.
Emphasize the result	Make the two measurements directly comparable while retaining individual lines and all observations.
Simplify it	Remove decorative backgrounds and repeated explanation; retain four objects, three relationships, and the legend.
Journal style	Use a white background and less decoration; apply the agreed text hierarchy and color roles.
Complete the legend	Draft what can be identified from the image; the author supplies sample sizes, conditions, and statistical methods.

These requests do not guarantee error-free execution. They make execution assessable. Image edits can affect the whole image, so even a request to change

one label should be followed by another check of objects and relationships.

2.8 Read the brief as if the research were unfamiliar

Temporarily set aside the background in your head and look for missing information in the text.

"The treatment group is higher" lacks a variable, condition, and evidential basis. "Split into two groups after this step" lacks a grouping rule. "Improves performance" lacks a metric, unit, and comparison. Supply the missing information or mark the claim as unresolved.

A small mapping table can help: statement, source location, visual representation, and question still to resolve. In real projects, identify a column, file version, or record page. In this exercise, identify a teaching assumption. This prevents poorly sourced claims from slipping into an image.

2.9 Exercise and acceptance criteria

Complete two briefs: one for your research and one for either the fictional mechanism or paired-data example. Retain the original description and write a short revision note explaining which relationships, sources, and acceptance criteria you added.

Check whether objects can be counted, relationships can be examined one by one, data constraints map to columns, unknowns remain explicit, and the output has a concrete use.

You now have an input that can move into figure creation. Chapter 3 will explore different compositions from the same brief and compare the work they ask the reader to do.

Appendix A

Book roadmap and companion materials

This edition contains drafts of the preface and Chapters 1 and 2. Later chapters follow research communication tasks. Each will cover AI assistance, a SciDraw AI workflow, author review, a revision example, and deliverables.

Ch.	Topic	Main deliverable
1	AI figure workflows	Brief, route, source archive; drafted
2	Research to drawing brief	Relationship table and acceptance criteria; drafted
3	Composition and sketches	Two candidates and a selection rationale
4	Visual language	Color, type, labels, and arrow conventions
5	Data visualization	Data, figure, revisions, and checks
6	Methods and workflows	Step table, diagram, and conditions
7	Mechanisms and structures	Relationship table, corrections, and legend
8	Experimental images	Originals, panel plan, processing record

Ch.	Topic	Main deliverable
9	Panels and legends	Assembled figure and reviewed legend
10	Graphical abstracts	Brief, layout candidates, final figure
11	Effective revisions	Issue list and reviewed iterations
12	Presentations and posters	Versions for different viewing situations
13	Checks and delivery	Figures, legends, sources, AI-use record

A.1 A reusable brief

```
[Audience and use]
[One-sentence purpose]
[Materials and sources]
[Required content]
[Relationships and order]
[Evidence: observation, interpretation, hypothesis, unknown]
[Visual plan]
[Constraints: what must not change]
[Delivery requirements and date checked]
[Acceptance criteria]
```

A.2 Simulated-data dictionary

`data/paired-example.csv` contains manually constructed teaching data. Eight fictional objects each have two measurements in arbitrary units. There are no missing values or predefined exclusion criteria. Seven objects increase; SIM04 decreases from 18 to 17. The before mean is 13.125 and the after mean is 16.5. These descriptions support no real research, causal, or treatment-effect claim.

`scripts/plot_paired.py` reads the original CSV, checks these properties, and renders the plots. It produces PDF and PNG figures with Chinese or English labels. It does not invent, remove, or alter observations.

A.3 Figure provenance and AI-use records

The mechanism image comes from Codex's built-in image generation. The executed prompt is in `provenance/mechanism-prompt.txt`; the review is in `mechanism-review.json` in the same directory. The tool did not expose a specific model name, so the book does not infer one. The original output is preserved without visual postprocessing.

The executed prompt moves explanations of relationships to the caption instead of placing words inside the image. This preserves their meaning and lets both editions share the figure. Review covers object counts, directions, line styles, and unsupported additions. Slight tonal variation within nodes is recorded as a style deviation. The fictional relationships have no status as biological evidence.

The workflow is a Codex-authored TikZ diagram, and interface images are real development screenshots. ASSETS.md lists sources and uses. Future real-research cases must add source material, any necessary permission, actual tool operations, and revision records.

Bibliography

[1] Nicolas P. Rougier, Michael Droettboom, and Philip E. Bourne. Ten Simple Rules for Better Figures. *PLOS Computational Biology*, 10(9):e1003833, 2014. https://doi.org/10.1371/journal.pcbi.1003833.

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